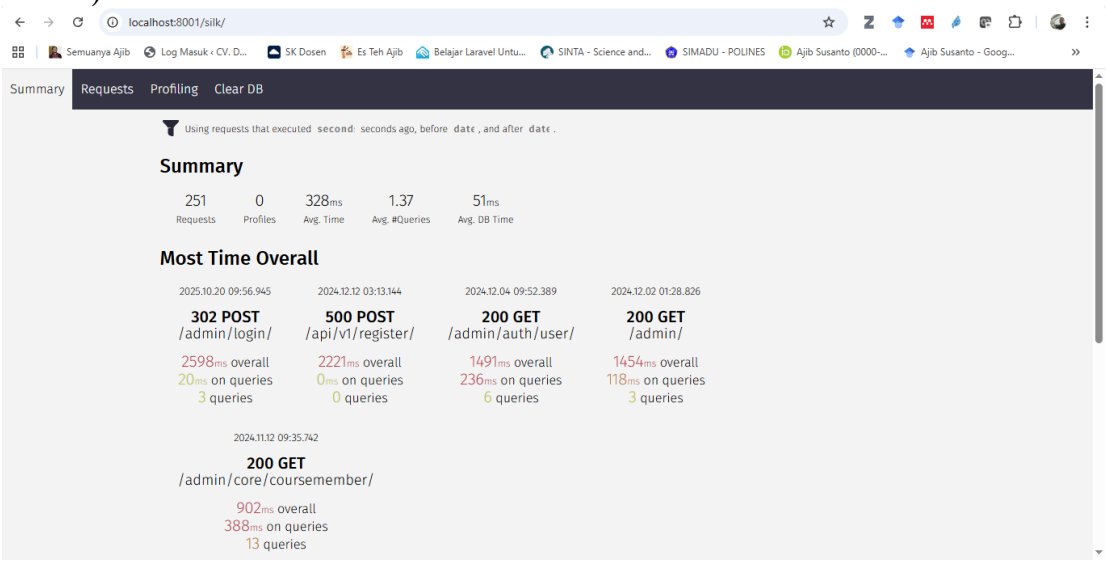
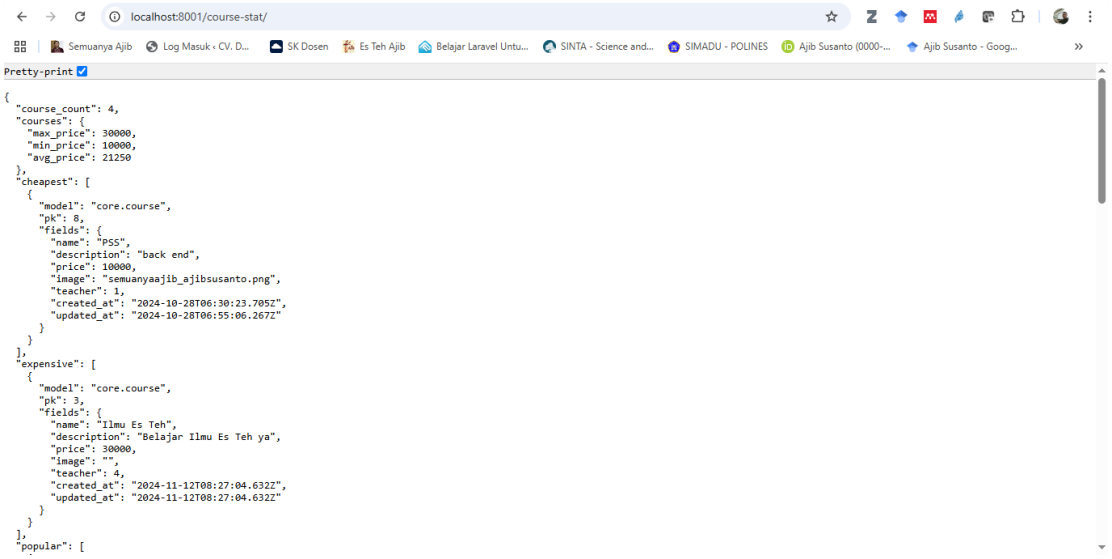
Pertemuan 7 : Profiling dan Optimasi ORM

1. Dari latihan sebelumnya Simple_LMS, tambahkan library Django silk (materi 7).

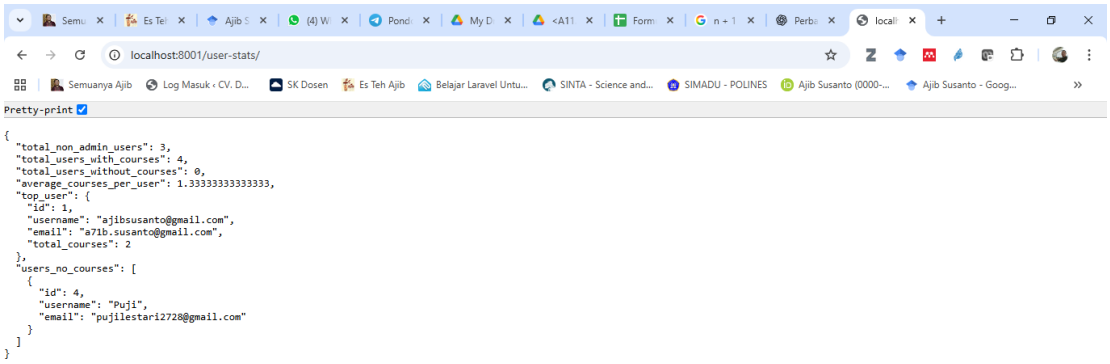


2. Eksekusi Query Relasi beberapa tabel pada materi 7.

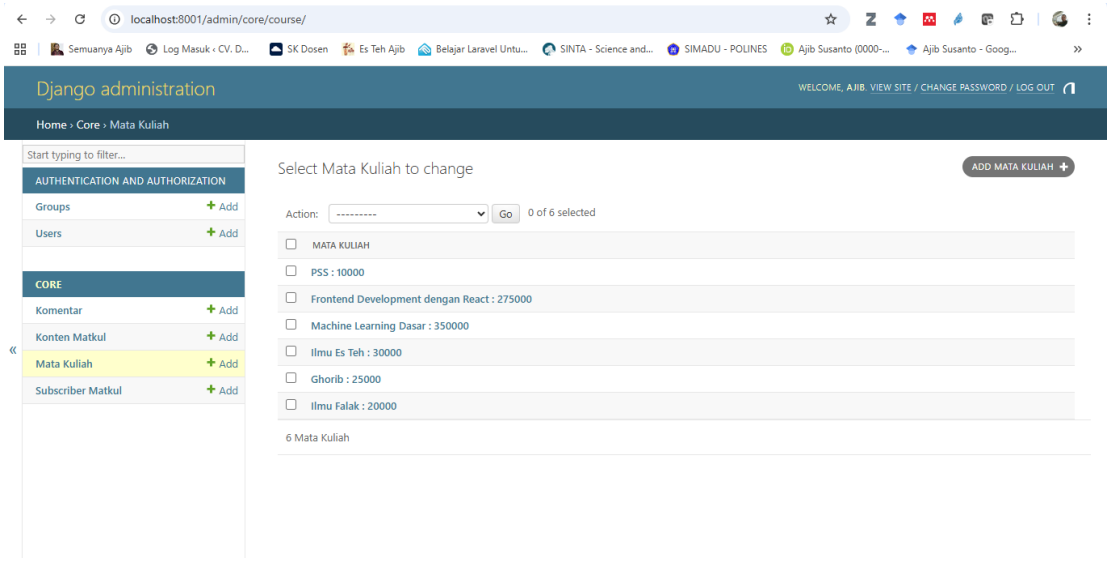
   a. Course Statistik



b. User Statistik



c. Halaman Admin



Langkah-langkahnya :

1. Tambahkan library django-silk di requirement.txt :

```
django
psycopg2-binary
pillow
django-silk
```

2. Di setting.py ubah konfigurasi MIDDLEWARE dan INSTALLED_APPS :

```
'silk',
```

Dan

```
'silk.middleware.SilkyMiddleware',
```

```python
INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
    'silk',
    'ninja_simple_jwt',
]

MIDDLEWARE = [
    'django.middleware.security.SecurityMiddleware',
    'django.contrib.sessions.middleware.SessionMiddleware',
    'django.middleware.common.CommonMiddleware',
    'django.middleware.csrf.CsrfViewMiddleware',
    'django.contrib.auth.middleware.AuthenticationMiddleware',
    'django.contrib.messages.middleware.MessageMiddleware',
    'django.middleware.clickjacking.XFrameOptionsMiddleware',
    'silk.middleware.SilkyMiddleware',
]
```

3. Di urls.py tambahkan :

```
urlpatterns += [path('silk/', include('silk.urls', namespace='silk'))]
```

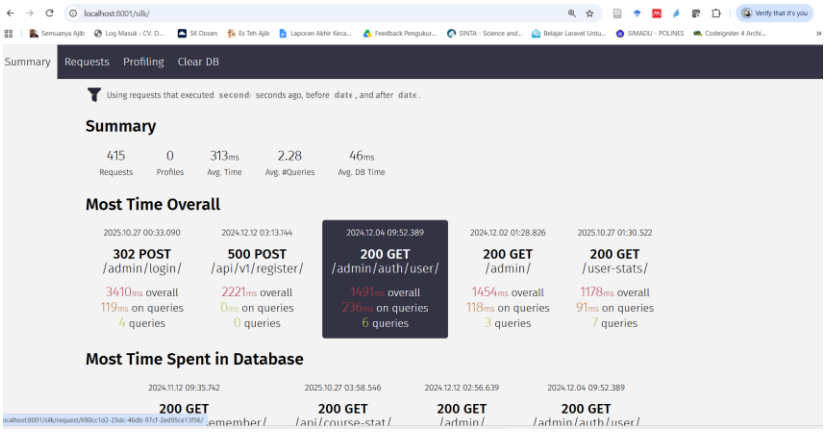
4. Rebuild image :

```
docker compose up -d –build
```

5. Migrasi table silk :

```
docker exec simple_lms python manage.py makemigrations
```

6. Cek hasil di : `localhost:8001/silk`



7. Tambahkan code di views.py :

```
def core(request):
   #return HttpResponse("Selamat Datang - e-LMS!")
   return render(request, "core/home.html")

def statistik(request):
    return render(request, "core/statistik.html")

def statistik_from_api(request):
    return render(request, "core/statistik_from_api.html")

def users(request):
  myusers = User.objects.all().values()
  template = loader.get_template('all_users.html')
  context = {
    'myusers': myusers,
  }
  return HttpResponse(template.render(context, request))

#User.objects.create_user(username="usertesting",
email="usertest@email.com", password="sanditesting")
#Course.objects.create(name=row['name'],
description=row['description'],                price=row['price'],
teacher=User.objects.get(pk=1))

def courseStat(request):
    # Ambil semua data course
    courses = Course.objects.all()

    # Hitung statistik harga
    stats = courses.aggregate(
        max_price=Max('price'),
        min_price=Min('price'),
        avg_price=Avg('price')
    )

    # Ambil course dengan harga termurah & termahal
    cheapest = Course.objects.filter(price=stats['min_price'])
    expensive = Course.objects.filter(price=stats['max_price'])

    # Ambil course terpopuler & tidak populer (berdasarkan jumlah
member)
    popular = (
        Course.objects
        .annotate(member_count=Count('coursemember'))
        .order_by('-member_count')[:5]
    )
    unpopular = (
        Course.objects
        .annotate(member_count=Count('coursemember'))
        .order_by('member_count')[:5]
    )

    # Hasil akhir
    result = {
        'course_count': courses.count(),
```

```
        'courses': stats,
        'cheapest': serializers.serialize('python', cheapest),
        'expensive': serializers.serialize('python', expensive),
        'popular': serializers.serialize('python', popular),
        'unpopular': serializers.serialize('python', unpopular),
    }

    return JsonResponse(result, safe=False)

def userStatistics(request):
    # 1 Semua user non-admin
        non_admin_users   =   User.objects.filter(is_staff=False,
is_superuser=False)
    total_non_admin_users = non_admin_users.count()

    # 2 User yang membuat course
                                users_with_courses              =
User.objects.filter(course__isnull=False).distinct()
    total_users_with_courses = users_with_courses.count()

    # 3 User tanpa course
    users_without_courses = non_admin_users.exclude(
        id__in=users_with_courses.values_list('id', flat=True)
    )
    total_users_without_courses = users_without_courses.count()

    # 4 Rata-rata jumlah course yang diikuti per user
    avg_courses_per_user = (
        CourseMember.objects.values('user_id')
        .annotate(total=Count('course_id'))
        .aggregate(avg_courses=Avg('total'))['avg_courses'] or 0
    )

    # 5 User yang mengikuti course terbanyak
    top_user = (
        CourseMember.objects.values('user_id')
        .annotate(total=Count('course_id'))
        .order_by('-total')
        .first()
    )

    top_user_data = {}
    if top_user:
                                        top_user_instance         =
User.objects.get(id=top_user['user_id'])
        top_user_data = {
            "id": top_user_instance.id,
            "username": top_user_instance.username,
            "email": top_user_instance.email,
            "total_courses": top_user['total']
        }

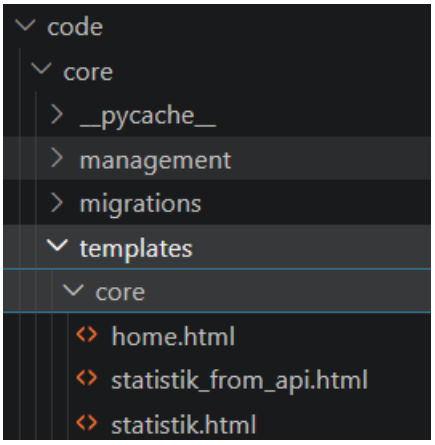
    # 6 Daftar user tanpa course sama sekali
    users_no_courses = non_admin_users.exclude(
                id__in=CourseMember.objects.values_list('user_id',
flat=True)
```

```
    )
    users_no_courses_list = list(
        users_no_courses.values('id', 'username', 'email')
    )

    # 7️⃣ Hasil akhir
    result = {
        "total_non_admin_users": total_non_admin_users,
        "total_users_with_courses": total_users_with_courses,
       "total_users_without_courses": total_users_without_courses,
        "average_courses_per_user": avg_courses_per_user,
        "top_user": top_user_data,
        "users_no_courses": users_no_courses_list,
    }

    return JsonResponse(result, safe=False)
```

8. Tambahkan code html di folder code/core/templates/core :



home.html :

```
<!doctype html>
<html lang="id">
<head>
  <meta charset="utf-8" />
    <meta   name="viewport"   content="width=device-width,initial-
scale=1" />
  <title>e-LMS • Selamat Datang</title>
  <style>
    :root{
           --bg:#0f1724;   --card:#0b1220;   --accent:#06b6d4;   --
muted:#94a3b8;
      --glass: rgba(255,255,255,0.04);
    }
    *{box-sizing:border-box}
    body{
        margin:0;  font-family:  Inter,  system-ui,  -apple-system,
"Segoe UI", Roboto, "Helvetica Neue", Arial;
      background: linear-gradient(180deg, #071024 0%, #071827 60%);
       color:#e6eef6; -webkit-font-smoothing:antialiased;
```

```
      -moz-osx-font-smoothing:grayscale;
      min-height:100vh; display:flex; align-items:center; justify-
content:center;
      padding:32px;
    }
    .container{
      width:100%; max-width:1100px;
       background: linear-gradient(180deg, rgba(255,255,255,0.02),
rgba(255,255,255,0.01));
      border-radius:14px; box-shadow: 0 10px 30px rgba(2,6,23,0.6);
        overflow:hidden; display:grid; grid-template-columns: 1fr
380px; gap:0;
    }

    /* left */
    .hero{
      padding:48px;
    }
    .logo{
       display:inline-block; padding:8px 12px; border-radius:10px;
background:var(--glass);
      font-weight:600; color:var(--accent); margin-bottom:18px;
    }
    h1{margin:0 0 12px; font-size:28px; line-height:1.05}
    p.lead{margin:0 0 22px; color:var(--muted); max-width:60ch}
      .features{display:flex; gap:12px; flex-wrap:wrap; margin-
top:22px}
     .chip{background:rgba(255,255,255,0.02); padding:10px 12px;
border-radius:10px; font-size:14px; color:#cfeefc}

    .actions{margin-top:28px; display:flex; gap:12px}
    .btn{
        padding:12px 16px; border-radius:10px; font-weight:600;
cursor:pointer; border:0;
      display:inline-flex; align-items:center; gap:8px;
    }
             .btn-primary{background:linear-gradient(90deg,var(--
accent),#7dd3fc); color:#042027; box-shadow: 0 6px 18px
rgba(6,182,212,0.12)}
         .btn-ghost{background:transparent; color:var(--muted);
border:1px solid rgba(255,255,255,0.04)}

    /* right card */
    .right{
              padding:24px; background:linear-gradient(180deg,
rgba(255,255,255,0.01), rgba(255,255,255,0.02));
      border-left: 1px solid rgba(255,255,255,0.03);
        display:flex; flex-direction:column; gap:18px; justify-
content:center;
    }
    .stat{
       display:flex; align-items:center; gap:12px; padding:14px;
background:rgba(255,255,255,0.02); border-radius:10px;
    }
      .stat .num{font-size:20px; font-weight:700; color:var(--
accent)}
    .stat small{display:block; color:var(--muted); font-size:13px}
```

```
      footer{padding:14px;  text-align:center;  color:var(--muted);
font-size:13px}
    @media (max-width:900px){
      .container{grid-template-columns:1fr; padding:0}
      .hero{padding:28px}
    }
  </style>
</head>
<body>
  <div class="container" role="main" aria-label="e-LMS Welcome">
    <div class="hero">
      <div class="logo">e-LMS</div>
         <h1>Selamat  Datang  di  e-LMS  —  Platform  Pembelajaran
Online</h1>
      <p class="lead">Kelola mata kuliah, unggah konten, dan terima
komentar siswa dengan mudah. Dashboard yang ringan, aman, dan siap
dipakai di lingkungan pendidikan.</p>

      <div class="features" aria-hidden="false">
        <div class="chip">Manajemen Kursus</div>
        <div class="chip">Konten Video & File</div>
        <div class="chip">Komentar & Diskusi</div>
         <div class="chip" onclick="location.href='/api/statistik-
api/'">Statistik Pengguna</div>
      </div>

      <div class="actions">
                           <button     class="btn     btn-primary"
onclick="location.href='/admin/'">Buka Admin</button>
                            <button     class="btn     btn-ghost"
onclick="location.href='/api/course-stat/'">Lihat
Statistik</button>
      </div>

         <div  style="margin-top:28px;  color:var(--muted);  font-
size:14px">
        <strong>Tips:</strong> klik <em>Buka Admin</em> untuk login
sebagai staff dan coba fitur manajemen course.
      </div>
    </div>

    <aside class="right" aria-label="Sekilas Statistik">
         <div  style="font-size:13px;  color:var(--muted)">Sekilas
Statistik</div>

      <div class="stat">
        <div>
          <div class="num" id="stat-courses">—</div>
          <small>Total Mata Kuliah</small>
        </div>
      </div>

      <div class="stat">
        <div>
          <div class="num" id="stat-users">—</div>
          <small>Total Pengguna</small>
```

```
        </div>
      </div>

      <div class="stat">
        <div>
          <div class="num" id="stat-comments">—</div>
          <small>Jumlah Komentar</small>
        </div>
      </div>

     <footer>e-LMS · dibuat untuk pembelajaran yang mudah</footer>
    </aside>
  </div>

  <script>
    // fetch ringkas stats (jika endpoint ada). Jika tidak ada,
tetap aman.
    (async function(){
      try{
        const res = await fetch('/api/course-stat/');
        if(!res.ok) return;
        const data = await res.json();
            document.getElementById('stat-courses').textContent  =
data.course_count ?? '0';
      }catch(e){/* ignore */}
      try{
        const res2 = await fetch('/api/user-stats/');
        if(res2.ok){
          const ud = await res2.json();
              document.getElementById('stat-users').textContent  =
ud.total_non_admin_users ?? '0';
        }
      }catch(e){}
      // komentar count fallback: could fetch an endpoint, or leave
—
    })();
  </script>
</body>
</html>
```

Statistic.html:

```
<!DOCTYPE html>
<html lang="id">
<head>
    <meta charset="UTF-8">
     <meta  name="viewport"  content="width=device-width,  initial-
scale=1.0">
    <title>Statistik e-LMS</title>
    <style>
        body {
            margin: 0;
            font-family: 'Poppins', sans-serif;
            background-color: #f3f4f6;
            color: #111827;
        }
```

```
        header {
            background: linear-gradient(135deg, #4f46e5, #3b82f6);
            color: white;
            padding: 30px;
            text-align: center;
            box-shadow: 0 4px 8px rgba(0,0,0,0.15);
        }

        header h1 {
            margin: 0;
            font-size: 2rem;
        }

        main {
            max-width: 1000px;
            margin: 40px auto;
            padding: 0 20px;
        }

        .cards {
            display: grid;
             grid-template-columns: repeat(auto-fit, minmax(220px,
1fr));
            gap: 20px;
        }

        .card {
            background-color: white;
            border-radius: 16px;
            box-shadow: 0 4px 10px rgba(0,0,0,0.1);
            padding: 20px;
            text-align: center;
            transition: transform 0.2s ease;
        }

        .card:hover {
            transform: translateY(-5px);
        }

        .card h2 {
            margin: 10px 0 5px;
            font-size: 2rem;
            color: #2563eb;
        }

        .card p {
            color: #6b7280;
            font-size: 0.95rem;
        }

        .chart {
            background-color: white;
            margin-top: 40px;
            border-radius: 16px;
            box-shadow: 0 4px 10px rgba(0,0,0,0.1);
            padding: 30px;
        }
```

```
        canvas {
            width: 100%;
            height: 300px;
        }

        footer {
            text-align: center;
            padding: 20px;
            color: #6b7280;
            font-size: 0.9rem;
        }
    </style>
</head>
<body>
    <header>
        <h1>📊 Statistik e-LMS</h1>
       <p>Ringkasan aktivitas pengguna dan konten pembelajaran</p>
    </header>

    <main>
        <section class="cards">
            <div class="card">
                <h2>120</h2>
                <p>Total Pengguna</p>
            </div>
            <div class="card">
                <h2>34</h2>
                <p>Kursus Aktif</p>
            </div>
            <div class="card">
                <h2>285</h2>
                <p>Materi Tersedia</p>
            </div>
            <div class="card">
                <h2>1,042</h2>
                <p>Komentar Dikirim</p>
            </div>
        </section>

        <section class="chart">
                 <h3  style="text-align:center;">Aktivitas  Belajar
Bulanan</h3>
            <canvas id="chartCanvas"></canvas>
        </section>
    </main>

    <footer>
        © 2025 e-LMS | Dikembangkan oleh Tim RPL
    </footer>

    <!-- Chart.js CDN -->
    <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
    <script>
                                        const        ctx        =
document.getElementById('chartCanvas').getContext('2d');
        new Chart(ctx, {
```

```
            type: 'line',
            data: {
                labels: ['Jan', 'Feb', 'Mar', 'Apr', 'Mei', 'Jun',
'Jul', 'Agu', 'Sep', 'Okt'],
                datasets: [{
                    label: 'Aktivitas Belajar',
                     data: [40, 55, 60, 75, 90, 80, 95, 100, 85,
120],
                    borderColor: '#2563eb',
                    backgroundColor: 'rgba(37,99,235,0.2)',
                    borderWidth: 3,
                    fill: true,
                    tension: 0.4,
                    pointRadius: 5
                }]
            },
            options: {
                responsive: true,
                plugins: {
                    legend: { display: false }
                },
                scales: {
                    y: { beginAtZero: true }
                }
            }
        });
    </script>
</body>
</html>
```

Statistic_from_api.html :

```
<!doctype html>
<html lang="id">
<head>
  <meta charset="utf-8" />
    <meta   name="viewport"   content="width=device-width,initial-
scale=1" />
  <title>Statistik • e-LMS</title>

  <style>
    :root{
      --bg:#f6f8fb; --card:#ffffff; --muted:#6b7280;
               --accent:#2563eb;    --accent-2:#06b6d4;    --glass:
rgba(10,15,30,0.03);
      --radius:12px;
    }
    *{box-sizing:border-box}
    body{
       margin:0; font-family: Inter, ui-sans-serif, system-ui, -
apple-system, "Segoe UI", Roboto, "Helvetica Neue", Arial;
     background: linear-gradient(180deg,#eef2ff 0%, #f8fafc 100%);
      color:#0f1724; -webkit-font-smoothing:antialiased; -moz-osx-
font-smoothing:grayscale;
      padding:28px;
    }
```

```
    .wrap{max-width:1100px; margin:0 auto;}
          header{display:flex;    align-items:center;    justify-
content:space-between; gap:16px; margin-bottom:20px}
    header h1{margin:0; font-size:20px; color:var(--accent)}
    header p{margin:0; color:var(--muted); font-size:14px}

    .grid{
      display:grid;
      grid-template-columns: repeat(4, 1fr);
      gap:16px;
    }
    @media (max-width:900px){
      .grid{grid-template-columns: repeat(2, 1fr)}
    }
    @media (max-width:520px){
      .grid{grid-template-columns: 1fr}
    }

    .card{
           background:var(--card);   border-radius:var(--radius);
padding:18px; box-shadow: 0 6px 20px rgba(15,23,42,0.06);
    }
    .card .label{font-size:13px; color:var(--muted)}
    .card .value{font-size:24px; font-weight:700; margin-top:6px;
color:#0b1220}

      .columns{display:grid;  grid-template-columns:  1fr  420px;
gap:16px; margin-top:20px}
    @media (max-width:900px){ .columns{grid-template-columns: 1fr;
} }

      .panel{background:var(--card);  border-radius:var(--radius);
padding:18px; box-shadow: 0 6px 20px rgba(15,23,42,0.06);}

     .list{display:flex;  flex-direction:column;  gap:8px;  margin-
top:10px}
         .list-item{display:flex;   justify-content:space-between;
gap:12px;   padding:10px;   border-radius:10px;   background:var(--
glass); align-items:center}
    .list-item .name{font-weight:600}
    .tiny{font-size:12px; color:var(--muted)}

      .section-title{display:flex;  justify-content:space-between;
align-items:center; gap:12px}
      .section-title  h3{margin:0;  font-size:16px;  color:var(--
accent)}
    .empty{color:var(--muted); padding:18px; text-align:center}

    /* table-like list for cheapest/expensive mobile friendly */
    .mini-table{width:100%; border-radius:10px; overflow:hidden}
         .mini-row{display:flex;   justify-content:space-between;
padding:10px 12px; background:transparent; border-bottom:1px solid
rgba(15,23,42,0.03)}
    .mini-row:last-child{border-bottom:0}

    footer{margin-top:22px; text-align:center; color:var(--muted);
font-size:13px}
```

```
  </style>
</head>
<body>
  <div class="wrap" role="main" aria-live="polite">
    <header>
      <div>
        <h1>📊 Statistik e-LMS</h1>
        <p>Data diambil dari <code>/course-stat/</code></p>
      </div>
      <div id="last-updated" class="tiny">—</div>
    </header>

    <section class="grid" aria-label="Ringkasan statistik">
      <div class="card">
        <div class="label">Total Mata Kuliah</div>
        <div class="value" id="stat-courses">—</div>
      </div>
      <div class="card">
        <div class="label">Harga Rata-rata</div>
        <div class="value" id="stat-avg">—</div>
      </div>
      <div class="card">
        <div class="label">Harga Terendah</div>
        <div class="value" id="stat-min">—</div>
      </div>
      <div class="card">
        <div class="label">Harga Tertinggi</div>
        <div class="value" id="stat-max">—</div>
      </div>
    </section>

    <div class="columns">
      <div>
        <div class="panel" aria-labelledby="popular-title">
          <div class="section-title"><h3 id="popular-title">Kursus
Terpopuler</h3><div class="tiny" id="popular-count">—</div></div>
          <div id="popular-list" class="list"></div>
        </div>

             <div  class="panel"  style="margin-top:16px"  aria-
labelledby="unpopular-title">
                  <div  class="section-title"><h3  id="unpopular-
title">Kursus Tidak Populer</h3></div>
          <div id="unpopular-list" class="list"></div>
        </div>
      </div>

      <aside>
        <div class="panel" aria-labelledby="chart-title">
          <div class="section-title">
            <h3 id="chart-title">Grafik: Peserta per Kursus</h3>
            <div class="tiny">Top 10</div>
          </div>
             <canvas  id="popularChart"  width="400"  height="240"
style="max-width:100%; margin-top:14px"></canvas>
        </div>
```

```
        <div class="panel" style="margin-top:16px">
                    <div   class="section-title"><h3>Termurah   &
Termahal</h3></div>
          <div class="mini-table" style="margin-top:10px">
            <div style="display:flex; gap:12px;">
                                                <div     style="flex:1;
padding:8px"><strong>Termurah</strong></div>
                                                <div     style="flex:1;
padding:8px"><strong>Termahal</strong></div>
            </div>
            <div style="display:flex; gap:12px">
                    <div  style="flex:1;  background:var(--glass);
padding:8px">
                <div id="cheapest-list"></div>
              </div>
                    <div  style="flex:1;  background:var(--glass);
padding:8px">
                <div id="expensive-list"></div>
              </div>
            </div>
          </div>
        </div>
      </aside>
    </div>

    <footer>
      <div id="status" class="tiny">Menunggu data…</div>
    </footer>
  </div>

  <!-- Chart.js (CDN) -->
  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
  <script>
    // Expected JSON shape (example)
    // {
    //   "course_count": 6,
                            //                              "stats":
{"max_price":1500000,"min_price":400000,"avg_price":983333.33},
              //            "cheapest":[{"id":5,"name":"HTML      &
CSS","price":"400000.00"}],
                //               "expensive":[{"id":3,"name":"Data
Science","price":"1500000.00"}],
               //              "popular":[{"id":3,"name":"Machine
Learning","price":"1500000.00","member_count":4}, ...],
    //   "unpopular":[...]
    // }

    const endpoint = '/course-stat/';

    // helper: friendly currency
    const fmt = (n) => {
      if (n === null || n === undefined) return '—';
      const x = Number(n);
      if (isNaN(x)) return String(n);
      return 'Rp ' + x.toLocaleString('id-ID');
    };
```

```
    // render functions
    function renderSummary(data){
           document.getElementById('stat-courses').textContent   =
data.course_count ?? '0';
             document.getElementById('stat-avg').textContent    =
fmt(data.stats?.avg_price);
             document.getElementById('stat-min').textContent    =
fmt(data.stats?.min_price);
             document.getElementById('stat-max').textContent    =
fmt(data.stats?.max_price);
    }

    function renderList(containerId, items, extraKey){
      const container = document.getElementById(containerId);
      container.innerHTML = '';
      if(!items || items.length === 0){
           container.innerHTML  =  `<div  class="empty">Tidak  ada
data</div>`;
        return;
      }
      items.forEach(it => {
        const div = document.createElement('div');
        div.className = 'list-item';
        div.innerHTML = `<div>
                 <div  class="name">${escapeHtml(it.name  ||  '(no
name)')}</div>
            <div class="tiny">ID: ${it.id}</div>
          </div>
          <div>
            <div class="tiny">${fmt(it.price)}</div>
              ${extraKey ? `<div class="tiny">× ${it[extraKey] ??
0}</div>` : ''}
          </div>`;
        container.appendChild(div);
      });
    }

    function renderMiniList(containerId, items){
      const el = document.getElementById(containerId);
      el.innerHTML = '';
        if(!items  ||  items.length  ===  0){  el.innerHTML  =  '<div
class="tiny">—</div>'; return; }
      items.forEach(it=>{
        const row = document.createElement('div');
        row.className = 'mini-row';
                                  row.innerHTML       =       `<div
style="flex:1">${escapeHtml(it.name)}</div><div        style="min-
width:90px;text-align:right">${fmt(it.price)}</div>`;
        el.appendChild(row);
      });
    }

    function escapeHtml(s){
               return    String(s).replace(/[&<>"']/g,     (m)=>({
'&':'&amp;','<':'&lt;','>':'&gt;','"':'&quot;',"'":'&#39;' }[m]));
    }
```

```
    // chart
    let chart = null;
    function drawChart(popular){
                                      const           ctx           =
document.getElementById('popularChart').getContext('2d');
      if(chart) chart.destroy();
      const top = (popular || []).slice(0,10);
      const labels = top.map(i=>i.name);
      const data = top.map(i=>Number(i.member_count || 0));
      chart = new Chart(ctx, {
        type: 'bar',
        data: {
          labels,
          datasets: [{
            label: 'Jumlah Member',
            data,
            backgroundColor: '#60a5fa',
            borderColor: '#2563eb',
            borderWidth: 1
          }]
        },
        options: {
          responsive: true,
          plugins: { legend: { display: false } },
          scales: {
            y: { beginAtZero: true, ticks: { stepSize: Math.max(1,
Math.ceil((Math.max(...data,1))/5)) } }
          }
        }
      });
    }

    // fetch and render
    async function loadStats(){
      const statusEl = document.getElementById('status');
      statusEl.textContent = 'Mengambil data dari server…';
      try{
        const res = await fetch(endpoint, {cache: "no-store"});
        if(!res.ok) throw new Error('HTTP ' + res.status);
        const data = await res.json();

        renderSummary(data);

        renderList('popular-list', data.popular, 'member_count');
                    renderList('unpopular-list',    data.unpopular,
'member_count');

        renderMiniList('cheapest-list', data.cheapest);
        renderMiniList('expensive-list', data.expensive);

        drawChart(data.popular);

           document.getElementById('popular-count').textContent  =
(data.popular?.length ?? 0) + ' results';
         document.getElementById('last-updated').textContent = new
Date().toLocaleString('id-ID');
```

```
        statusEl.textContent = 'Data berhasil dimuat.';
      }catch(err){
        console.error('Gagal memuat statistik:', err);
       statusEl.textContent = 'Gagal memuat data — periksa endpoint
/course-stat/.';
        // fallback: tampil contoh statik minimal agar UI tidak
kosong
        const fallback = {
          course_count: 0,
           stats: { max_price: null, min_price: null, avg_price:
null },
          cheapest: [],
          expensive: [],
          popular: [],
          unpopular: []
        };
        renderSummary(fallback);
        renderList('popular-list', [], 'member_count');
        renderList('unpopular-list', [], 'member_count');
        renderMiniList('cheapest-list', []);
        renderMiniList('expensive-list', []);
        drawChart([]);
      }
    }

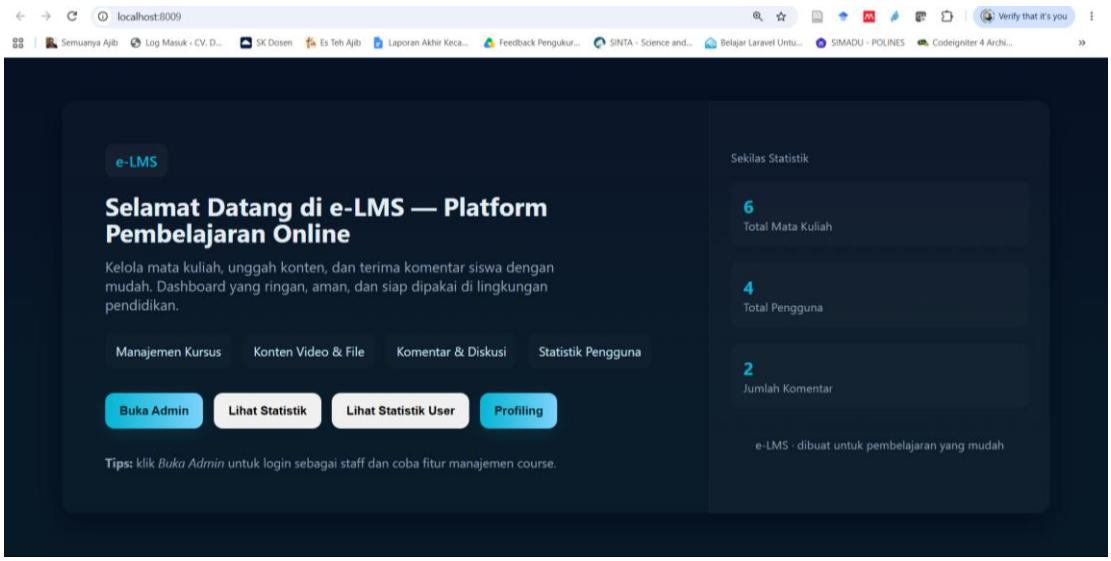
    // auto load
    loadStats();
  </script>
</body>
</html>
```

9. Update dan sesuikan urls.py :

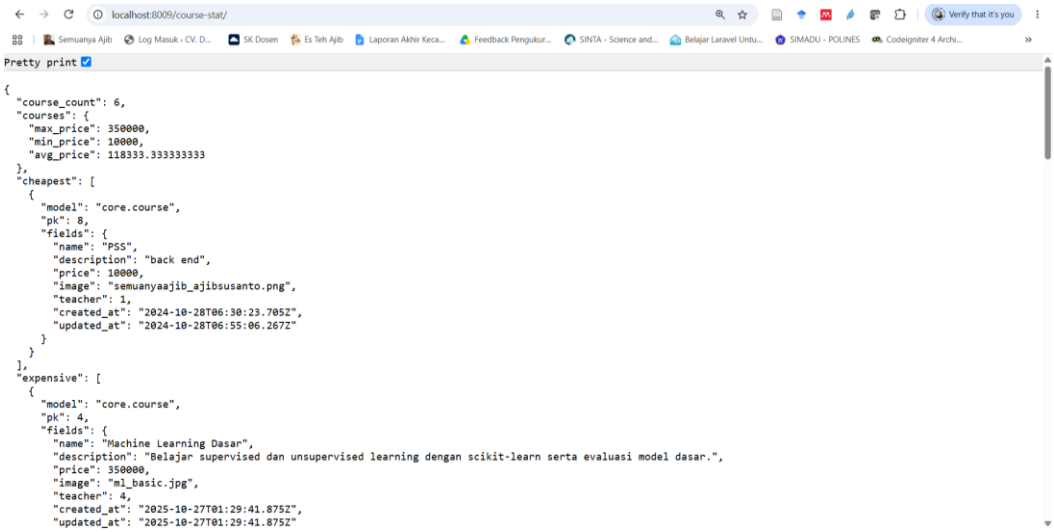
```
from django.urls import path
from . import views

urlpatterns = [
    path('', views.core, name='core'),
    path('core/', views.core, name='core'),
    path('users/', views.core, name='users'),
    path('statistik/', views.statistik, name='statistik'),
    path('statistik-api/', views.statistik_from_api, name='statistik_api'),
    path('course-stat/', views.courseStat, name='course_stat'),
    path('user-stats/', views.userStatistics, name='userStatistics'),
]
```

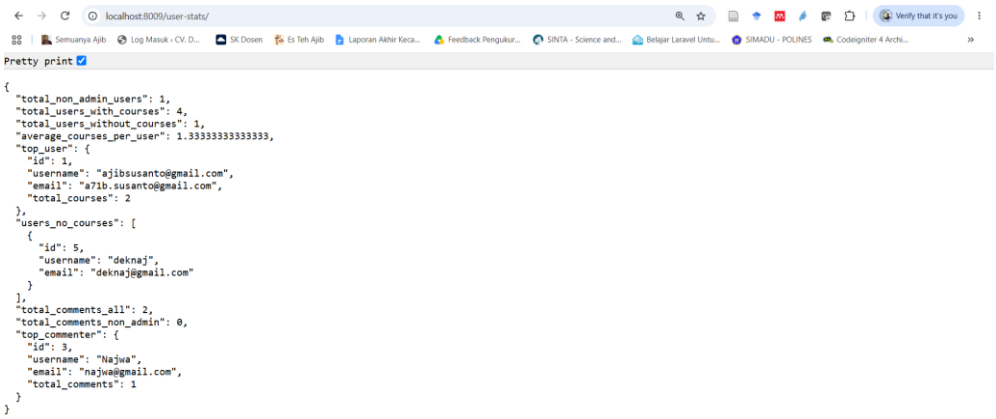
10. Lakukan `docker compose down` dan `docker compose up -d --build`
11. Cek hasilnya di **docker** atau **localhost:8001**



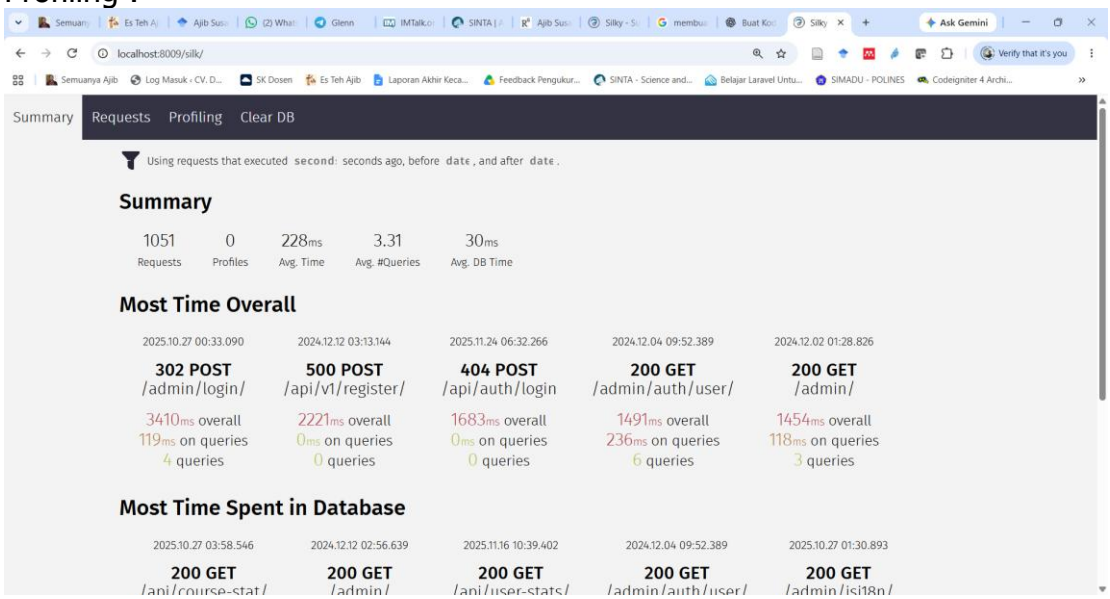
Lihat Statistik :



Lihat Statistik Pengguna :



Profiling :



Statistik Pengguna :

